CLOSED IMMERSIONS APPEAR AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1896-1908, following the statement at 1886-1894.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Affineness appears at a finite stage. lemma-limit-is-affine (same chapter, card L26293, cited at ~1899-1900). V_i x_{Y_i} X_i is affine for some i.
- Scheme-level descent of closed immersions. Limits, Lemma limits-lemma-descend-closed-immersion-finite-presentation (cross-chapter, cited at ~1902-1903). V_i x_{Y_i} X_i -> V_i closed immersion for i large.
- Local detection of integral morphisms. Morphisms of Spaces, Lemma spaces-morphisms-lemma-integral-local (cross-chapter, cited at ~1904-1906). f_i closed immersion detected on the etale cover.

Source reference. Original lemma 07SQ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1370–1370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZP: focused-seeds/geometry/S9-spaces-limits-descend-closed-immersion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ZQ: focused-seeds/geometry/S9-spaces-limits-descend-closed-immersion/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084W. If*

(1) *f is a closed immersion,*

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *f_0 is locally of finite type,*

then f_i is a closed immersion for some $i \geq 0$.

Proof. Choose an affine scheme V_0 and a surjective étale morphism $V_0 \to Y_0$. Set $V_i = V_0 \times_{Y_0} Y_i$ and $V = V_0 \times_{Y_0} Y$. Since f is a closed immersion we see that $V \times_Y X = \lim V_i \times_{Y_i} X_i$ is a closed subscheme of the affine scheme V. By Lemma 07SQ we see that $V_i \times_{Y_i} X_i$ is affine for some $i \geq 0$. Increasing i if necessary we find that $V_i \times_{Y_i} X_i \to V_i$ is a closed immersion by Limits, Lemma 01ZP. For this i the morphism f_i is a closed immersion (Morphisms of Spaces, Lemma 03ZQ). $\square$